

GEOTWINAI

AI-Powered Digital Twin for Smart Cities

Nagpur Urban Planning & Decision Support Report

Study Area	Nagpur, Maharashtra
Satellite Dataset	Sentinel-2 Level-2A
Satellite Source	Microsoft Planetary Computer
Analysis Resolution	10 metres
Scene Date	2026-05-12

GeoTwinAI integrates geospatial datasets, satellite-derived indicators and machine learning to support urban planning and smart-city decision-making.

1. Executive Summary

GeoTwinAI is an AI-powered geospatial decision-support platform developed for urban planning and smart-city analysis in Nagpur, Maharashtra. The system integrates infrastructure datasets with Sentinel-2 satellite observations to provide a spatial understanding of the urban environment. The selected satellite scene has a recorded cloud cover of 0.00% and a scene date of 2026-05-12. The platform analyses vegetation, built-up areas and water-related characteristics using NDVI, NDBI and NDWI. Infrastructure datasets include buildings, hospitals, schools, parks, roads and water bodies. The final system combines GIS visualization, machine learning and Power BI analytics into a unified urban planning workflow.

2. Project Objectives

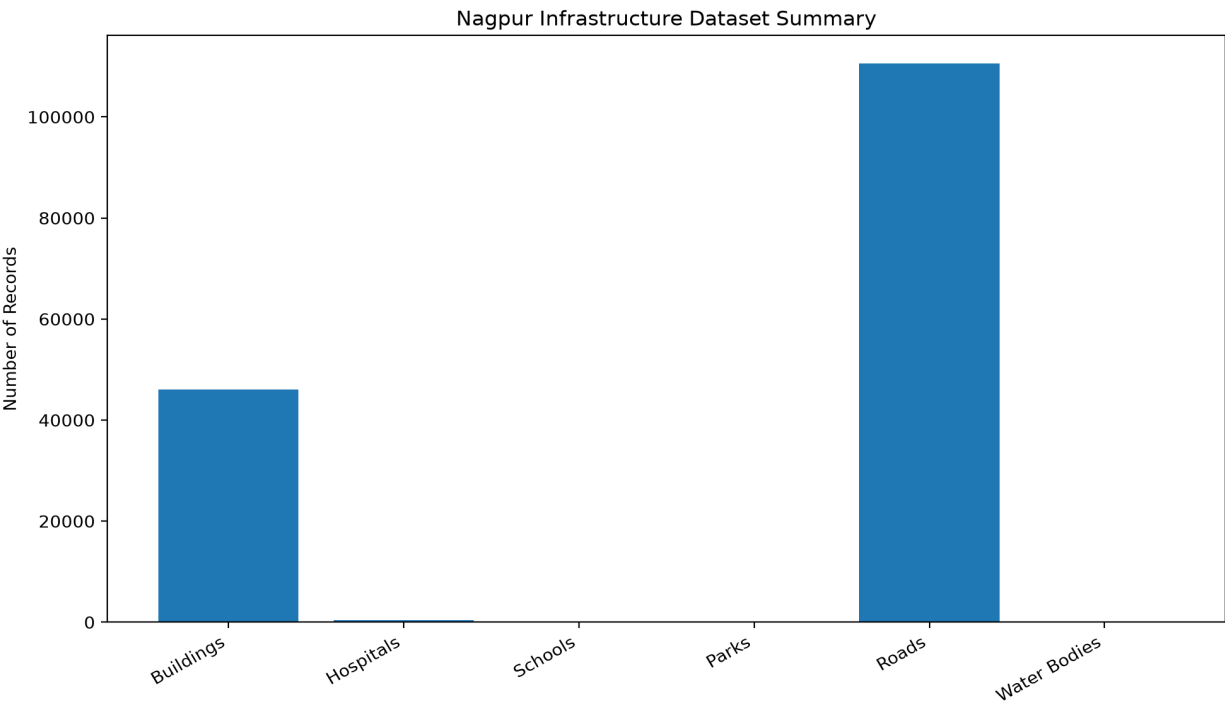
- Develop an AI-powered Digital Twin framework for Nagpur.
- Integrate satellite and GIS datasets.
- Analyse vegetation and environmental conditions.
- Identify built-up and water-related characteristics.
- Map important urban infrastructure.
- Create machine-learning-based urban priority analysis.
- Provide interactive GIS visualization.
- Provide analytical dashboards through Power BI.
- Support data-driven urban planning decisions.

3. Data Sources

Dataset	Purpose
Buildings	Urban building and infrastructure analysis
Hospitals	Healthcare infrastructure mapping
Schools	Educational infrastructure mapping
Parks	Green and open-space analysis
Roads	Transportation network analysis
Water Bodies	Water-resource analysis
Sentinel-2	Satellite and environmental analysis
Machine Learning	Urban priority prediction
Power BI	Interactive analytical dashboard

4. Urban Infrastructure Statistics

Infrastructure Type	Records
Buildings	46052
Hospitals	412
Schools	75
Parks	142
Roads	110566
Water Bodies	0

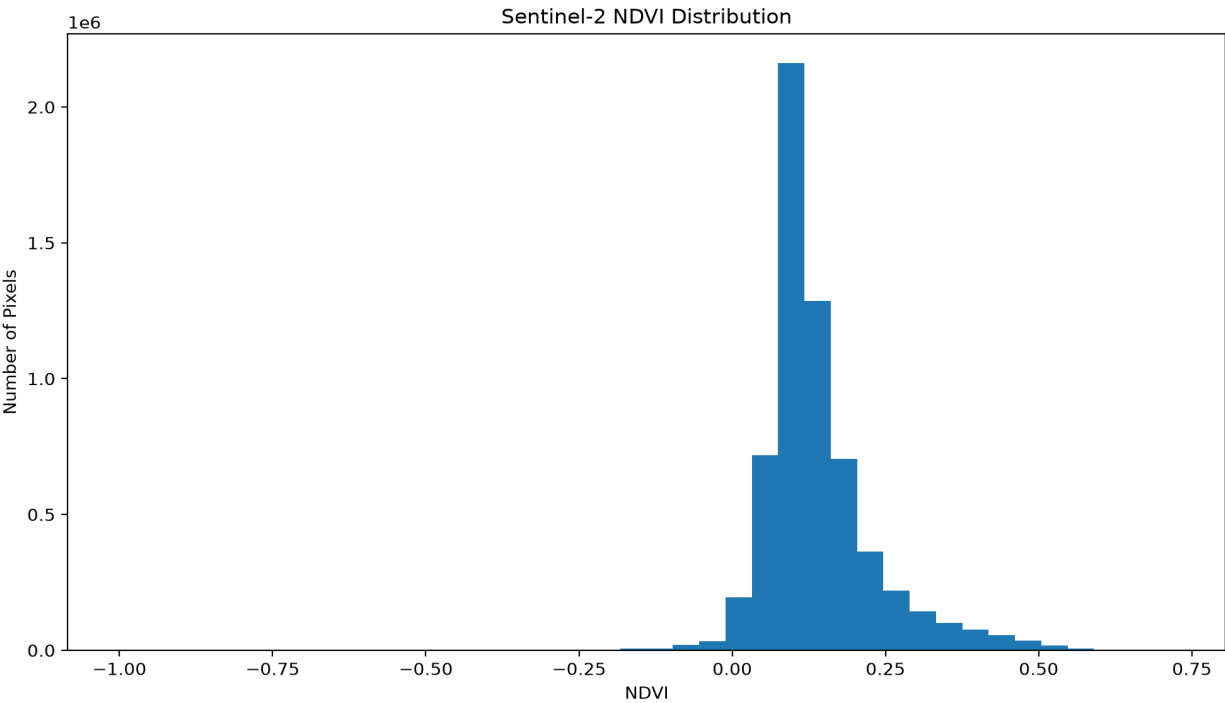


5. Sentinel-2 Satellite Analysis

The satellite analysis uses Sentinel-2 Level-2A imagery obtained through Microsoft Planetary Computer. The selected scene date is 2026-05-12, with a reported cloud cover of 0.00%. A common 10-metre spatial reference grid is used for integrated multi-band analysis. The processed dataset includes B01, B02, B03, B04, B05, B06, B07, B08, B8A, B09, B11 and B12.

6. Spectral Indices

Index	Purpose	Mean
NDVI	Vegetation condition	0.1392
NDBI	Built-up area indication	0.0804
NDWI	Water-related analysis	-0.1982



7. Sentinel-2 All-Band Analysis

The all-band Sentinel-2 dataset preserves spectral information required for environmental and urban analysis. Processed bands include: B01 - Coastal/Aerosol B02 - Blue B03 - Green B04 - Red B05 - Vegetation Red Edge B06 - Vegetation Red Edge B07 - Vegetation Red Edge B08 - Near Infrared B8A - Narrow Near Infrared B09 - Water Vapour B11 - Short-Wave Infrared B12 - Short-Wave Infrared These bands support vegetation, water, soil, built-up and land-cover analysis.

8. Machine Learning Analysis

The machine-learning component of GeoTwinAI is designed to classify urban areas according to planning priority. The feature-engineering stage combines infrastructure and environmental attributes into a machine-learning dataset. The trained model can generate urban priority predictions such as High, Medium and Low. These predictions can support identification of areas requiring additional infrastructure or environmental attention. Model performance should be evaluated using appropriate classification metrics such as accuracy, precision, recall and F1-score.

9. GeoTwinAI Workflow

1. Data collection
2. Data cleaning
3. Satellite processing
4. Spectral index generation
5. Feature engineering
6. Machine learning
7. Prediction
8. Interactive GIS visualization
9. Power BI dashboard
10. Urban planning decision support

10. Key Findings

- The processed infrastructure datasets contain 46052 building records.
- The healthcare dataset contains 412 hospital records.
- The education dataset contains 75 school records.
- The parks dataset contains 142 park records.
- The water-body dataset contains 0 records.
- The selected Sentinel-2 scene has 0.00% reported cloud cover.
- The mean NDVI is 0.1392.
- The mean NDBI is 0.0804.
- The mean NDWI is -0.1982.

11. Recommendations

- Use the interactive GIS map to inspect spatial distribution of infrastructure.
- Use NDVI to identify vegetation-rich and vegetation-deficient areas.
- Use NDBI to identify highly built-up areas.
- Use NDWI and water-body data to support water-resource planning.
- Use ML priority classifications to identify areas requiring planning attention.
- Use Power BI for executive-level monitoring and comparative analysis.
- Periodically update Sentinel-2 observations to monitor urban environmental change.
- Integrate official MRSAC datasets wherever available for production-level analysis.

12. Limitations

- The Nagpur bounding box represents an analysis area and is not necessarily a formal administrative boundary.
- Satellite observations are affected by atmospheric and seasonal conditions.
- Cloud-cover metadata is scene-level and does not necessarily represent cloud conditions at every pixel.
- The current analysis uses one selected best-available scene rather than a full temporal time series.
- Machine-learning predictions depend on the quality and representativeness of the training dataset.
- Infrastructure records should be validated against authoritative datasets before operational deployment.

13. Conclusion

GeoTwinAI provides a unified framework for combining geospatial infrastructure data, satellite remote sensing and machine learning for smart-city planning. The integration of Sentinel-2 spectral information, environmental indices, infrastructure datasets and machine-learning predictions provides a spatially oriented view of urban conditions. The interactive GIS map provides detailed spatial exploration, while Power BI provides higher-level analytical dashboards for decision-makers. Future development can include multi-date satellite monitoring, improved administrative boundaries, official MRSAC datasets, advanced machine-learning models and automated data updates. The overall system can therefore serve as a foundation for an AI-enabled urban planning and decision-support platform.

14. Technical Information

Programming	Python
Satellite	Sentinel-2
Product	Sentinel-2 Level-2A
Satellite Source	Microsoft Planetary Computer
Spatial Reference Grid	10 metres
Indices	NDVI, NDBI, NDWI
GIS Visualization	Interactive HTML Map
Dashboard	Microsoft Power BI
Machine Learning	Urban Priority Classification
Report Format	PDF